

# **DHRUVA - Acronyms & Full Forms**

Here is the complete list of acronyms and their full forms used in the DHRUVA Landslide Early Warning System project:

## **General & Domain Terms:**

- SIH: Smart India Hackathon
- NER: North Eastern Region
- AI: Artificial Intelligence
- IoT: Internet of Things
- FoS: Factor of Safety

## **Government & Institutional Bodies:**

- IMD: India Meteorological Department
- CWC: Central Water Commission
- GSI: Geological Survey of India
- NDMA: National Disaster Management Authority

## **Technology & Software:**

- GIS: Geographic Information System
- PWA: Progressive Web App
- XGBoost: eXtreme Gradient Boosting (Machine Learning algorithm)
- LSTM: Long Short-Term Memory (Recurrent Neural Network architecture)

## **Sensors & Geospatial Data:**

- InSAR: Interferometric Synthetic Aperture Radar
- DEM: Digital Elevation Model
- TDR: Time-Domain Reflectometry (used for soil moisture sensing)
- GNSS: Global Navigation Satellite System (used for ground displacement)
- MEMS: Micro-Electromechanical Systems (used in micro-seismometers)